

①

Bio test - 3

Date _____
Page _____

Section - A

Question - 1

Papaver somniferum

Question - 2

Asymptote

Question - 3

Nucleotide

Question - 4

Question - 5

Progesterone

Question - 6

Two closely related species competing for the same limited resources cannot coexist, and the competitively inferior one will eventually be eliminated.

Question - 7

Cu-7

Question-8

It guides the entry of the pollen tube into the synergid.

Question-9

-

Question-10

IgA

Question-11

National Aids Control organisation

Question-12

Aneuploidy

Question-13

Rheumatoid Arthritis

Question-14

Tapetum

Question-15

Phototropy

Section - B

Question-16

Azoospermia

Oligospermia

• Complete absence of sperm in semen

• Presence of very low amount of sperm in semen

Question-17

Test Cross

Back cross

• Cross b/w an individual of heterozygous dominant genotype and its recessive parent.

• Cross b/w an F_1 gen and its homozygous dominant parent.

Question-18

Trichoderma acts as an effective biocontrol agents against soil borne plant pathogens.

Question-19

Mammalian forelimbs exhibit divergent evolution, through homologous structures, sharing a common internal skeletal structure that adapted to perform different functions.



Question-20

Benign

- Stick to their original location and don't spread to other body parts

Malignant

- Invade and damage surrounding tissues, displaying metastasis.

Question-21

- Supply of oxygen and nutrients to the developing fetus
- Acts as an endocrine tissue, secreting hCG, hPL, oestrogen and progesterone, which are essential.

Question-22 ♥♥♥

Amniocentesis - Diagnostic procedure where amniotic fluid is aspirated, centrifuged, cells are cultured and analysed for karyotypic abnormalities.

It is used to detect genetic disorders in the developing fetus.

Question-23

- Cultured easily due to their short life cycle.
- A single mating produces a large no. of offsprings with clear phenotypic sexual dimorphism.

Question-24

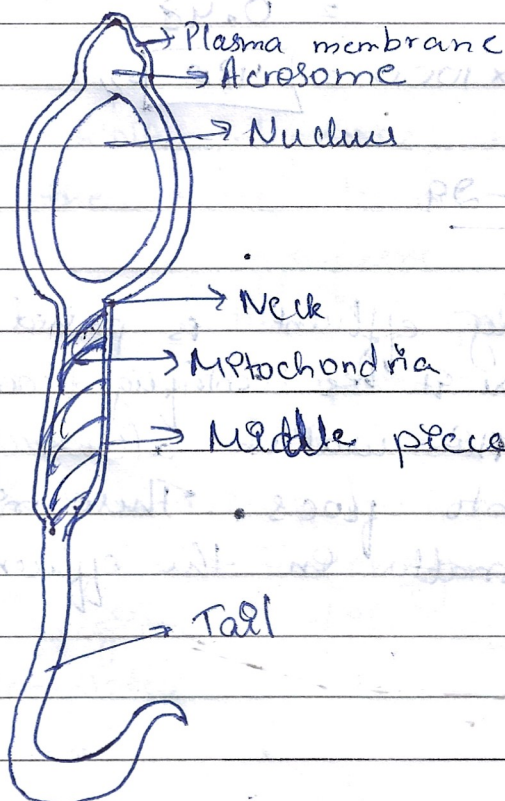
- Pollen grains are light, non-sticky and easy to travel far with the wind.
- They are produced in large quantities.

Question-25

Max population size of a species that a particular environment can support over time sustainably.

Section-C

Question-26



Question-27

GnRH - Acts on the anterior pituitary gland to stimulate LH & FSH

LH - Acts on Leydig cells to stimulate androgen secretion, in turn stimulating spermatogenesis

FSH - Acts on Sertoli cells, stimulating inhibin, stimulating spermiogenesis

Question-28

$$a + b = 1$$

$$a = 1 - 0.4$$

$$a = 0.6$$

$$2ab = 2 \times 0.4 \times 0.6$$

$$= 0.48$$

$$0.48 \times 1000 = \boxed{480} \text{ p.p.m.}$$

Question-29

The primary effluent is passed into large aeration tanks where it is aerated and air is pumped into it. This aeration allows growth of aerobic microbes into flocs. These microbes consume organic matter in the effluent, decreasing the BOD.

Question - 30

- Physical barriers : Prevent entry of pathogens
Eg: Skin
- Physiological barriers: Inhibit microbial growth through chemicals
Eg: HCl in stomach

Question - 31

The physical connection of 2 or more genes located on the same chromosome, so that they are inherited together is linkage.

Question - 32

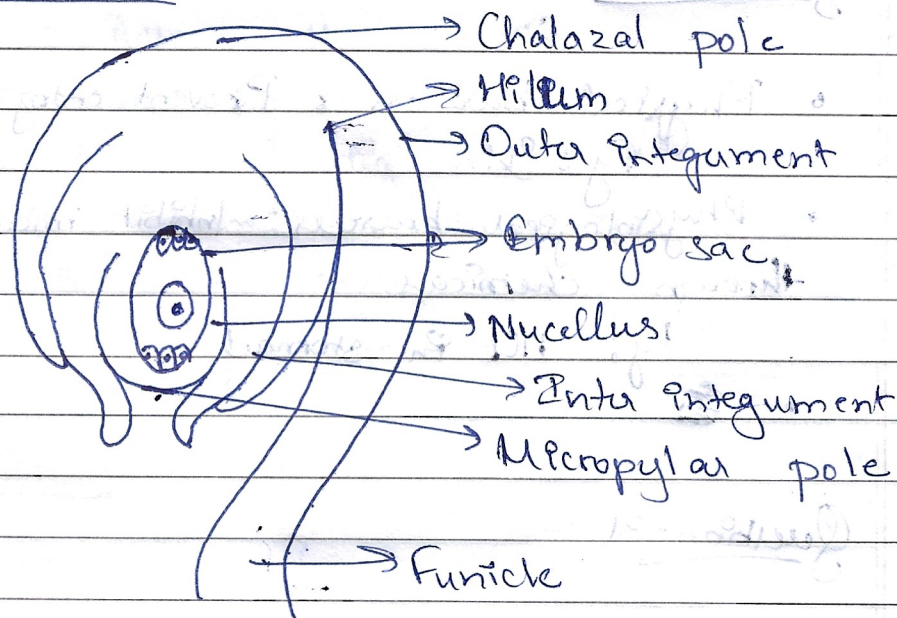
Initiation: RNA polymerase binds to the promoter region of the gene to unwind the DNA and start its synthesis.

Elongation: The main RNA polymerase enzyme moves down the template strand and synthesise a complementary RNA strand in the 5'-3' direction using nucleoside triphosphate.

Termination: When the enzyme reaches the terminator region, it dissociates from the DNA template.

⑧

Question - 33



Question - 34

Active

- Produced actively by the host's ~~body~~ own immune system due to exposure to antigens
- Slow onset
- Eg: Immunity after a natural recovery from chickenpox

Passive

- Initiated by the direct transfer of preformed antibodies into a non-immune individual.
- Short lived
- Eg: Colostrum providing IgA to a newborn.

Question - 35

The wasp carries pollen from its natal fig and cross-pollinates the florets of another fig. This provides a site for the female wasp to lay eggs within the ovary of the flower, developing wasp larva on a portion of the developing seeds inside the fig fruit. This is mutualism.

9

Section - D

36

Option - 1

- (a) Gases used: CH_4 , NH_3 , H_2 and H_2O
Ratio: $\text{CH}_4 : \text{NH}_3 : \text{H}_2$ in a 2 : 1 : 2 ratio
Temp: 800°C
Continuous high-voltage electric discharge through tungsten electrodes connected to a condenser that circulates cooled liquid water.
- (b) It confirmed the Haldane - Oparin hypothesis of chemical evolution.

~~27~~

~~37~~

38

Option - 2

- (a) Process of Translation:
- 1) Activation & Charging of tRNA: Amino acids are activated in the presence of ATP and attached to their cognate tRNA by aminoacyl-tRNA synthetase enzymes.
 - 2) Initiation: 30S binds to mRNA at the start codon (AUG). The initiator formyl-methionyl-tRNA pairs with AUG, after which 50S joins to form 70S initiation complex.

3) Elongation: Charged tRNAs enter the matching ribosomal A-site codon-anticodon triplets.

Peptidyl transferase catalyses peptide bond formation b/w adjacent amino acids. The ribosome moves down the mRNA one codon at a time from A-site to P-site.

4) Termination: When a stop codon enters the A-site, a release factor binds to it, causing the release of the synthesised ~~poly peptide~~ polypeptide chain and dissociation of the ribosomal complex.

(b) UTRs are segments of mRNA that are not translated into protein, located at 5' & 3' ends. They are essential for post-transcriptional regulation, enhancing the efficiency of the translation process, maintaining mRNA stability and ensuring proper ribosome binding.

37 (Option-1)

- 1) Multiplication & growth phase: The germ cells differentiate into Oogonia through mitosis. Oogonia enter Meiosis I & arrest at Prophase I, becoming primary oocytes⁽²ⁿ⁾. These get surrounded by granulosa cell layers to form Primary Follicles, which becomes Secondary Follicles, into Tertiary follicles.
- 2) Maturation phase: Just before ovulation, the primary oocyte completes Meiosis I inside Graafian follicle. Forms a large haploid Secondary oocyte⁽ⁿ⁾ & a tiny 1st polar body⁽ⁿ⁾. Oocyte begins Meiosis II & arrests at Metaphase II. Due to the LH surge, the Graafian follicle ruptures & releases the secondary oocyte⁽ⁿ⁾ into fallopian tube. Completion of Meiosis II occurs only if sperm penetrates the secondary oocyte, releasing a Second Polar Body & forming a mature Ovum⁽ⁿ⁾.